SQL Study Plan Day 4

Programming Questions

Departments

DepartmentID (PK)	DepartmentName
1	IT
2	HR
3	Finance
4	Marketing
5	Sales
6	Operations

Employees

EmployeeID (PK)	EmployeeName	DepartmentID (FK)	ManagerID (FK)	Salary
101	Amit	1	NULL	90000
102	Priya	1	101	75000
103	Rahul	1	101	65000
104	Sneha	2	NULL	70000
105	Neha	2	104	50000
106	Karan	3	NULL	85000
107	Pooja	3	106	60000
108	Rohan	4	NULL	55000
109	Meera	1	101	75000
110	Arjun	5	NULL	80000
111	Kavita	5	110	65000
112	Dev	3	106	85000

Customers

CustomerID (PK)	CustomerName	City
201	Ananya	Pune
202	Vikram	Mumbai
203	Isha	Delhi
204	Arjun	Pune
205	Kavya	Chennai
206	Riya	Mumbai
207	Sameer	Delhi
208	Nisha	Chennai

Orders

OrderID (PK)	CustomerID (FK)	EmployeeID (FK)	OrderDate	TotalAmount
301	201	102	2026-01-10	12000
302	201	103	2026-02-15	18000
303	202	102	2026-03-05	25000
304	203	107	2026-03-20	30000
305	204	109	2026-04-12	15000
306	202	103	2026-05-18	22000
307	205	107	2026-06-10	40000
308	201	102	2026-07-22	10000
309	204	109	2026-08-01	28000
310	203	107	2026-08-15	35000
311	206	110	2026-08-20	45000
312	207	111	2026-09-01	20000
313	205	112	2026-09-05	32000
314	206	102	2026-09-10	18000
315	201	109	2026-09-15	27000
316	208	110	2026-09-20	50000
317	202	103	2026-10-01	35000
318	203	112	2026-10-05	22000

OrderDetails

OrderDetailID (PK)	OrderID (FK)	ProductName	Category	Quantity	UnitPrice
401	301	Laptop	Electronics	1	10000
402	301	Mouse	Accessories	2	1000
403	302	Monitor	Electronics	2	9000
404	303	Keyboard	Accessories	5	5000
405	304	Desk	Furniture	3	10000
406	305	Chair	Furniture	5	3000
407	306	Laptop	Electronics	2	11000
408	307	Phone	Electronics	4	10000
409	308	Mouse	Accessories	5	2000
410	309	Desk	Furniture	2	14000
411	310	Chair	Furniture	5	7000
412	311	Laptop	Electronics	3	15000
413	312	Table	Furniture	2	10000
414	313	Phone	Electronics	2	16000
415	314	Keyboard	Accessories	3	6000
416	315	Chair	Furniture	4	7000
417	316	Monitor	Electronics	5	10000
418	317	Laptop	Electronics	2	17500
419	318	Desk	Furniture	2	11000

1. Write a SQL query to find employees who earn more than the average salary of their department and have handled orders for at least 2 distinct customers whose combined order value exceeds 50,000. Display employee name, department name, salary, distinct customer count, and total order value.

2. Write a SQL query to find the top 2 highest-paid employees in each department, including ties, without using window functions. Exclude employees whose salary is below the department's average salary. Display department name, employee name, and salary.

3. Write a SQL query to identify customers whose total spending is greater than the average total spending of customers from the same city. Include only customers who have placed at least 3 orders and purchased products from at least 2 distinct categories. Display customer name, city, order count, category count, and total spending.

4. Write a SQL query to find departments whose total employee salary exceeds the average total salary expenditure across all departments that have employees. Include departments with no employees in the final output and display their salary expenditure as zero.

5. Write a SQL query to find employees who have handled orders for every customer who has placed at least one order. Display employee name, department name, and the number of distinct customers served. Do not use window functions.

6. Write a SQL query to identify customers who have placed orders in at least 3 different months and whose spending in each of those months exceeded the average monthly spending per customer across all customers. Display customer name, qualifying month count, and total spending.

7. Write a SQL query to find employees whose salary is greater than the salaries of all employees in at least one other department, but lower than the highest salary in their own department. Display employee name, department name, salary, and the highest salary in their department.

8. Write a SQL query to find product categories whose total revenue is greater than the average revenue per category. For each qualifying category, display its total revenue, number of distinct customers who purchased from it, and the employee who handled the highest-value order in that category. Include ties for the highest-value order.

9. Write a SQL query to identify customers who have purchased products from every category represented in the OrderDetails table and whose total spending exceeds the average total spending of customers who have purchased from all represented categories. Display customer name, distinct category count, and total spending.

10. Write a SQL query to find the employees who handled the highest total order value within each department, including ties. Only consider employees who handled orders for at least 2 distinct customers. Display department name, employee name, distinct customer count, total order value, and the difference between the employee's total order value and the department's average employee order value. Include departments with no qualifying employees.

Puzzles

1. You have 12 coins, one counterfeit, and a balance scale. The counterfeit may be heavier or lighter. Why does each weighing need to be planned to distinguish both coin identity and weight direction?

2. Nine balls look identical; one is heavier. Find it with a balance scale in two weighings.

3. A chocolate bar has 12 equal pieces. If pieces may be stacked between cuts, what is the minimum number of straight cuts needed to separate all pieces?

4. There are 25 horses and only five can race at a time. With no stopwatch, what is the minimum number of races needed to identify the three fastest horses?

5. A chessboard has two opposite corners removed. Can the remaining squares be covered by dominoes that each cover two adjacent squares? Explain using square colours.

Aptitude Questions

1. A committee of 5 is selected from 7 men and 6 women. In how many ways can it be formed with at least 3 women?

A) 525 B) 636 C) 686 D) 756

2. How many ways can 4 different prizes be distributed among 6 students if no student receives more than one prize?

A) 120 B) 240 C) 360 D) 720

3. Two cards are drawn from a deck without replacement. What is the probability that both are aces?

A) 1/221 B) 1/169 C) 1/52 D) 2/221

4. A family has two children. Assuming boys and girls are equally likely, what is the probability both are girls given that at least one is a girl?

A) 1/4 B) 1/3 C) 1/2 D) 2/3

5. A box contains 3 red, 5 blue and 2 green balls. What is the probability that a randomly selected ball is neither red nor green?

A) 1/5 B) 1/2 C) 3/5 D) 2/3

6. A 60-L solution contains acid and water in ratio 5:1. How much water must be added to make the ratio 5:2?

A) 8 L B) 10 L C) 12 L D) 15 L

7. A 25-L mixture of milk and water is in ratio 4:1. How much milk should be added so that the ratio becomes 5:1?

A) 4 L B) 5 L C) 6 L D) 7 L

8. A 50-L mixture contains 60% alcohol. How much pure alcohol should be added to make the concentration 70%?

A) 12.5 L B) 15 L C) 16⅔ L D) 20 L

9. M is the brother of N. N is the daughter of P. Q is the wife of P. R is the son of Q. How is R related to M?

A) Brother B) Son C) Nephew D) Cousin

10. Arun's mother has only one daughter, who is Arun's sister. Pointing to a photograph, Arun says: 'The man in the photograph is the father of her only son.' How is the man related to Arun?

A) Father B) Brother C) Brother-in-law D) Uncle